Airbnb Grands Project

Problem Statement

Airbnb Grands wants multiple five-star hotels across India. They have been in the hospitality industry for the past 20 years. Due to strategic moves from other competitors and ineffective decision-making, their occupancy rate has dropped to 60%. The company has hired a new CEO who has decided that the company should focus on the luxury segment. The company has hired a consultant, Prabhakar and Daily Intelligence*, to report their market share and revenue. Moreover, they do not have an in-house data analytics team to provide them with these insights.

Data Exploration and Data Cleaning

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_percentage_error
```

Overview Of Datasets

```
!ls -lrtsh data/*
```

```
total 1.8M
 4K -rw-r--r-- 1 root root  1.3K Dec 13 13:51 data/hotel_bookings.csv
 4K -rw-r--r-- 1 root root  1.3K Dec 13 13:51 data/hotel_bookings_1.csv
 4K -rw-r--r-- 1 root root  1.3K Dec 13 13:51 data/hotel_bookings_2.csv
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 4K -rw-r--r-- 1 root root  1.3K Dec 13 13:51 data/hotel_bookings_98.csv
 4K -rw-r--r-- 1 root root  1.3K Dec 13 13:51 data/hotel_bookings_99.csv
 4K -rw-r--r-- 1 root root  1.3K Dec 13 13:51 data/hotel_bookings_100.csv
```

Overview Of Columns

```
!t data/*100.csv
```

```
hotel,lead_time,arrival_date_month,stays_in_weekend_nights,stays_in_week_nights,
City Hotel,342,July,0,2,
City Hotel,500,July,0,4,
City Hotel,240,July,0,4,
City Hotel,133,July,0,0,
City Hotel,132,July,0,1,
City Hotel,132,July,0,1,
City Hotel,132,July,0,1
```

We should offer weekday discounts to increase bookings during low-demand days

We can introduce corporate booking offers to attract business travelers

We should provide long-day deals, where customers pay less per day when they stay longer

We can give early booking discounts to encourage advance reservations

We should create combo offers, such as room with breakfast or room with spa services

We should offer loyalty rewards to encourage repeat customers

We should provide online-only offers to increase bookings through digital platforms